SE Lab 3

Problem Statement 22, Smart Parking System: Component Diagram and Architecture Justification

Sunidhi Shekar
PES1UG24AM410

Component Diagram

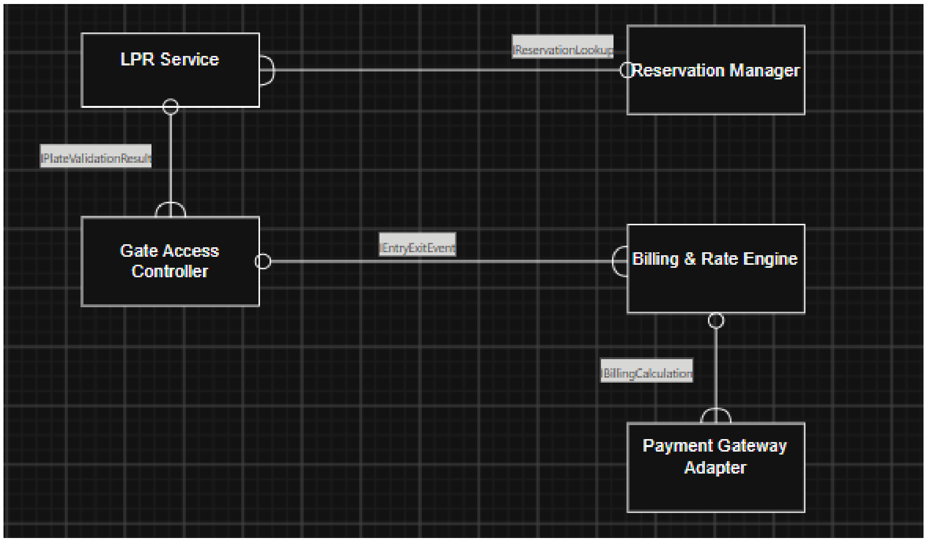

Figure 1: UML component diagram for the Smart Parking system.

Architecture Justification

Microservices Architecture was chosen for the Smart Parking Space Rental and License Validator System.

Architectural Choice

We picked Microservices Architecture. The system is split into small services: Reservation Manager, LPR Service, Gate Access Controller, Billing and Rate Engine, and Payment Gateway Adapter.

Two Reasons

1. Each service can grow on its own. The LPR Service gets busy scans during rush hours. The Billing and Rate Engine has a steady, lower load. Microservices let each one scale by itself instead of forcing one big system to handle the busiest part.

2. If one service fails, the rest keep working. If the Payment Gateway Adapter has a problem, for example if the outside payment company is slow, the gate should still open and close for cars. Because each part is separate, the Gate Access Controller keeps working even if billing or payment is having trouble.

Security Advantage

Each service only shares what it needs to share. The Gate Access Controller only gets a pass or fail result from the LPR Service. It never sees reservation records or payment details. This keeps sensitive data such as card details and reservation history safe inside their own services.

Performance Benefit

The LPR Service and Gate Access Controller are small and talk to each other directly. This is what makes the under 300 millisecond goal possible. Billing and warden alerts run after the gate has already opened, so slower tasks never block the fast, important path.